

L1.8 Linearization & Taylor

Work in your groups. Answers are in the document you get at the end — don't wait for it.

Linearization

1. Find the linearization of $f(x) = \frac{2}{\sqrt{x^2 - 5}}$ at $a = 3$.
2. Use it to estimate $f(3.1)$. Then say whether your estimate is too big or too small, and how you know **without** computing $f(3.1)$.
3. Estimate $(2.001)^5$ without a calculator.
4. A function is **concave down** everywhere. Sketch it with a tangent line drawn at some point a , and decide: is $L(x)$ an over-estimate or an under-estimate of $f(x)$? Does your answer depend on which side of a you are on?

Differentials and actual change

5. Find the differential dy .

(a) $y = \frac{1 + 2x}{1 + 3x}$

(b) $y = x^2 \sin(2x)$

6. For $y = x^3$ at $x = 2$ with $\Delta x = dx = 0.1$:

(a) Find Δy .

(b) Find dy .

(c) Which one did the tangent line give you, and what is the difference between them?

Error propagation

Stop. An angle is about to be multiplied by a length. Which measure does the angle have to be in — and what is the reason, not just the rule?

7. One side of a right triangle is known to be 20 cm long, and the angle opposite it is measured as 30° , with a possible error of 1° .

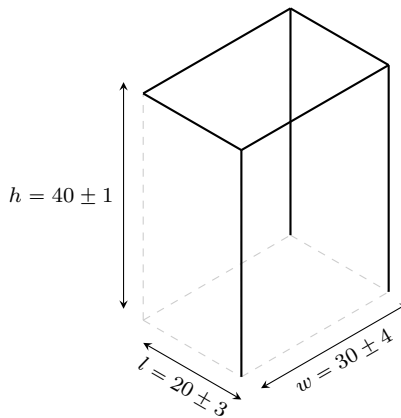
(a) Use differentials to estimate the error in the computed length of the hypotenuse.

(b) What is the percentage error?

8. A sphere's radius is measured to within 2%. To within what percentage is its volume known? Its surface area?

Stop. You know what a fractional uncertainty does to r^3 — one length, raised to a power. The box below is lwh : three *different* lengths, each with its own uncertainty. What do you think happens?

9. The length, width and height of a box are each known only to some uncertainty: $(l, w, h) = (20 \pm 3 \text{ cm}, 30 \pm 4 \text{ cm}, 40 \pm 1 \text{ cm})$.



- (a) What is the fractional uncertainty of each dimension?
- (b) What is the fractional uncertainty of the volume?
- (c) What is the uncertainty of the volume?

Summation notation and the closed forms

10. Evaluate.

(a) $\sum_{i=1}^4 (i^2 + 1)$ — write out every term.

(b) $\sum_{i=1}^{30} i^3$ — use a closed form.

(c) Check that $\sum_{i=1}^4 i^3$ really does equal $\left(\sum_{i=1}^4 i\right)^2$.

Maclaurin and Taylor series

11. Maclaurin series for $f(x) = \frac{1}{2-x}$.

n	$f^{(n)}(x)$	$f^{(n)}(0)$	$\frac{f^{(n)}(0)}{n!}$
0			
1			
2			
3			

$T_3(x) =$ _____

12. Maclaurin series for $f(x) = \frac{1}{1+x^2}$.

n	$f^{(n)}(x)$	$f^{(n)}(0)$	$\frac{f^{(n)}(0)}{n!}$
0			
1			
2			
3			

$T_3(x) =$ _____

13. Taylor series for $f(x) = \sqrt{x}$, centred at $a = 4$.

n	$f^{(n)}(x)$	$f^{(n)}(4)$	$\frac{f^{(n)}(4)}{n!}$
0			
1			
2			
3			

$T_3(x) =$ _____

Powers of $(x - 4)$, **not** powers of x .

14. **BOARDS** Estimate $\sqrt{10}$ without a calculator. Get as many decimal places as you can **defend** — be ready to say why you trust each one.